Rounded approximation codes for higher inductive-inductive Cauchy reals

The Univalent Foundations Program

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Editorial note. The complete fixed author proof is retained, including its explicit symmetry treatment of the final eight left-triangle cases. No separate target is counted for subsequent encode-decode or completeness consequences.

Editorial note. Native notation slips are retained and annotated:1194 uses the universally available relation at the roundedness margin theta;1225 has the second predicate at v as determined by1050 and1226;1276 the second conclusion is the approximation relation as determined by the immediately stated inductive triangle inequality and1277. These identify symbols using adjacent displayed rules and do not change the statement or add a proof premise.

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Editorial note. 2026-10-11 材料收口：复用既有 H2、来源许可和近邻判断；核对原命题与作者证明起止、所指上下文及装配范围齐备。未重证作者结论；沿用既有非重复依据。

Allowed standard prerequisites (editorial).

- Univalent type theory and supplied Cauchy-real recursion. Rational order, triangle inequality and positive epsilon refinements; dependent products and subset types; propositional truncation, function extensionality and univalence for mere propositions; the supplied joint Cauchy-real/closeness recursion and mere-property induction principles. Rounded code relations and their module properties are not allowed as inputs.

Definition 11.3.1. Let $\mathbb{R}_{\mathsf{c}}$ and the relation $\sim : \mathbb{R}_{\mathsf{c}} \times \mathbb{R}_{\mathsf{c}} \times \mathbb{Q}_{+} \to \mathcal{U}$ be the following higher inductive-inductive type family. The type $\mathbb{R}_{\mathsf{c}}$ of **Cauchy reals** is generated by the following constructors:

- *rational points:* for any $q : \mathbb{Q}$ there is a real $\mathsf{rat}(q)$.
- *limit points*: for any $x : \mathbb{Q}_+ \to \mathbb{R}_\mathsf{c}$ such that

$$\forall(\delta, \epsilon : \mathbb{Q}_+).\, x_\delta \sim_{\delta+\epsilon} x_\epsilon \tag{11.3.2}$$

  there is a point $\mathsf{lim}(x) : \mathbb{R}_\mathsf{c}$. We call x a **Cauchy approximation**.
- *paths:* for $u, v : \mathbb{R}_\mathsf{c}$ such that

$$\forall(\epsilon : \mathbb{Q}_+).\, u \sim_\epsilon v \tag{11.3.3}$$

  then there is a path $\mathsf{eq}_{\mathbb{R}_\mathsf{c}}(u, v) : u =_{\mathbb{R}_\mathsf{c}} v$.

Simultaneously, the type family $\sim : \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c} \to \mathbb{Q}_+ \to \mathcal{U}$ is generated by the following constructors. Here q and r denote rational numbers; δ, ϵ, and η denote positive rationals; u and v denote Cauchy reals; and x and y denote Cauchy approximations:

- for any q, r, ϵ, if $-\epsilon < q - r < \epsilon$, then $\mathsf{rat}(q) \sim_\epsilon \mathsf{rat}(r)$,
- for any q, y, ϵ, δ, if $\mathsf{rat}(q) \sim_{\epsilon-\delta} y_\delta$, then $\mathsf{rat}(q) \sim_\epsilon \mathsf{lim}(y)$,
- for any x, r, ϵ, δ, if $x_\delta \sim_{\epsilon-\delta} \mathsf{rat}(r)$, then $\mathsf{lim}(x) \sim_\epsilon \mathsf{rat}(r)$,
- for any $x, y, \epsilon, \delta, \eta$, if $x_\delta \sim_{\epsilon-\delta-\eta} y_\eta$, then $\mathsf{lim}(x) \sim_\epsilon \mathsf{lim}(y)$,
- for any u, v, ϵ, if $\xi, \zeta : u \sim_\epsilon v$, then $\xi = \zeta$ (propositional truncation).

The general induction principle, which we may call $(\mathbb{R}_\mathsf{c}, \sim)$**-induction**, is therefore a sort of joint $\mathbb{R}_\mathsf{c}$-induction and $\sim$-induction. Consider, for instance, its non-dependent version, which we call $(\mathbb{R}_\mathsf{c}, \sim)$**-recursion**, which is the one that we will have the most use for. Ordinary $\mathbb{R}_\mathsf{c}$-recursion tells us that to define a function $f : \mathbb{R}_\mathsf{c} \to A$ it suffices to:

(i) for every $q : \mathbb{Q}$ construct $f(\mathsf{rat}(q)) : A$,
(ii) for every Cauchy approximation $x : \mathbb{Q}_+ \to \mathbb{R}_\mathsf{c}$, construct $f(x) : A$, assuming that $f(x_\epsilon)$ has already been defined for all $\epsilon : \mathbb{Q}_+$,
(iii) prove $f(u) = f(v)$ for all $u, v : \mathbb{R}_\mathsf{c}$ satisfying $\forall(\epsilon : \mathbb{Q}_+).\, u \sim_\epsilon v$.

However, it is generally quite difficult to show (iii) without knowing something about how f acts on ϵ-close Cauchy reals. The enhanced principle of $(\mathbb{R}_\mathsf{c}, \sim)$-recursion remedies this deficiency, allowing us to specify an *arbitrary* "way in which f acts on ϵ-close Cauchy reals", which we can then prove to be the case by a simultaneous induction with the definition of f. This is the family of relations $\frown$. Since A is independent of $\mathbb{R}_\mathsf{c}$, we may assume for simplicity that $\frown$ depends only on A and $\mathbb{Q}_+$, and thus there is no ambiguity in writing $a \frown_\epsilon b$ instead of $(u, a) \frown_\epsilon (v, b)$. In this case, defining a function $f : \mathbb{R}_\mathsf{c} \to A$ by $(\mathbb{R}_\mathsf{c}, \sim)$-recursion requires the following cases (which we now write using the pattern-matching convention).

- For every $q : \mathbb{Q}$, construct $f(\mathsf{rat}(q)) : A$.
- For every Cauchy approximation $x : \mathbb{Q}_+ \to \mathbb{R}_\mathsf{c}$, construct $f(\mathsf{lim}(x)) : A$, assuming inductively that $f(x_\epsilon)$ has already been defined for all $\epsilon : \mathbb{Q}_+$ and form a "Cauchy approximation with respect to $\frown$", i.e. that $\forall(\epsilon, \delta : \mathbb{Q}_+).\, (f(x_\epsilon) \frown_{\epsilon+\delta} f(x_\delta))$.
- Prove that the relations $\frown$ are *separated*, i.e. that, for any $a, b : A$, $(\forall(\epsilon : \mathbb{Q}_+).\, a \frown_\epsilon b) \Rightarrow (a = b)$.
- Prove that if $-\epsilon < q - r < \epsilon$ for $q, r : \mathbb{Q}$, then $f(\mathsf{rat}(q)) \frown_\epsilon f(\mathsf{rat}(r))$.
- For any $q : \mathbb{Q}$ and any Cauchy approximation y, prove that $f(\mathsf{rat}(q)) \frown_\epsilon f(\mathsf{lim}(y))$, assuming inductively that $\mathsf{rat}(q) \sim_{\epsilon-\delta} y_\delta$ and $f(\mathsf{rat}(q)) \frown_{\epsilon-\delta} f(y_\delta)$ for some $\delta : \mathbb{Q}_+$, and that $\eta \mapsto f(x_\eta)$ is a Cauchy approximation with respect to $\frown$.
- For any Cauchy approximation x and any $r : \mathbb{Q}$, prove that $f(\mathsf{lim}(x)) \frown_\epsilon f(\mathsf{rat}(r))$, assuming inductively that $x_\delta \sim_{\epsilon-\delta} \mathsf{rat}(r)$ and $f(x_\delta) \frown_{\epsilon-\delta} f(\mathsf{rat}(r))$ for some $\delta : \mathbb{Q}_+$, and that $\eta \mapsto f(x_\eta)$ is a Cauchy approximation with respect to $\frown$.
- For any Cauchy approximations x, y, prove that $f(\mathsf{lim}(x)) \frown_\epsilon f(\mathsf{lim}(y))$, assuming inductively that $x_\delta \sim_{\epsilon-\delta-\eta} y_\eta$ and $f(x_\delta) \frown_{\epsilon-\delta-\eta} f(y_\eta)$ for some $\delta, \eta : \mathbb{Q}_+$, and that $\theta \mapsto f(x_\theta)$ and $\theta \mapsto f(y_\theta)$ are Cauchy approximations with respect to $\frown$.

Note that in the last four proofs, we are free to use the specific definitions of $f(\mathsf{rat}(q))$ and $f(\mathsf{lim}(x))$ given in the first two data. However, the proof of separatedness must apply to *any* two elements of A, without any relation to f: it is a sort of "admissibility" condition on the family of relations $\frown$. Thus, we often verify it first, immediately after defining $\frown$, before going on to define $f(\mathsf{rat}(q))$ and $f(\mathsf{lim}(x))$.

Under the above hypotheses, $(\mathbb{R}_\mathsf{c}, \sim)$-recursion yields a function $f : \mathbb{R}_\mathsf{c} \to A$ such that $f(\mathsf{rat}(q))$ and $f(\mathsf{lim}(x))$ are judgmentally equal to the definitions given for them in the first two clauses. Moreover, we may also conclude

$$\forall(u, v : \mathbb{R}_\mathsf{c}).\, \forall(\epsilon : \mathbb{Q}_+).\, (u \sim_\epsilon v) \to (f(u) \frown_\epsilon f(v)). \tag{11.3.4}$$

Theorem 11.3.5. *There is a family of mere relations* $\approx : \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c} \to \mathbb{Q}_+ \to$ Prop *such that*

$$(\mathsf{rat}(q) \approx_\epsilon \mathsf{rat}(r)) :\equiv (-\epsilon < q - r < \epsilon) \tag{11.3.6}$$

$$(\mathsf{rat}(q) \approx_\epsilon \mathsf{lim}(y)) :\equiv \exists(\delta : \mathbb{Q}_+).\, \mathsf{rat}(q) \approx_{\epsilon-\delta} y_\delta \tag{11.3.7}$$

$$(\mathsf{lim}(x) \approx_\epsilon \mathsf{rat}(r)) :\equiv \exists(\delta : \mathbb{Q}_+).\, x_\delta \approx_{\epsilon-\delta} \mathsf{rat}(r) \tag{11.3.8}$$

$$(\mathsf{lim}(x) \approx_\epsilon \mathsf{lim}(y)) :\equiv \exists(\delta, \eta : \mathbb{Q}_+).\, x_\delta \approx_{\epsilon-\delta-\eta} y_\eta. \tag{11.3.9}$$

Moreover, we have

$$(u \approx_\epsilon v) \Leftrightarrow \exists(\theta : \mathbb{Q}_+).\, (u \approx_{\epsilon-\theta} v) \tag{11.3.10}$$

$$(u \approx_\epsilon v) \to (v \sim_\delta w) \to (u \approx_{\epsilon+\delta} w) \tag{11.3.11}$$

$$(u \sim_\epsilon v) \to (v \approx_\delta w) \to (u \approx_{\epsilon+\delta} w). \tag{11.3.12}$$

The additional conditions (11.3.10)–(11.3.12) turn out to be required in order to make the inductive definition go through. Condition (11.3.10) is called being **rounded**. Reading it from right to left gives **monotonicity** of $\approx$,

$$(\delta < \epsilon) \wedge (u \approx_\delta v) \Rightarrow (u \approx_\epsilon v)$$

while reading it left to right to **openness** of $\approx$,

$$(u \approx_\epsilon v) \Rightarrow \exists(\delta : \mathbb{Q}_+).\, (\delta < \epsilon) \wedge (u \approx_\delta v).$$

Conditions (11.3.11) and (11.3.12) are forms of the triangle inequality, which say that $\approx$ is a "module" over $\sim$ on both sides.

证明. We will define $\approx : \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c} \to \mathbb{Q}_+ \to \mathsf{Prop}$ by double $(\mathbb{R}_\mathsf{c}, \sim)$-recursion. First we will apply $(\mathbb{R}_\mathsf{c}, \sim)$-recursion with codomain the subset of $\mathbb{R}_\mathsf{c} \to \mathbb{Q}_+ \to \mathsf{Prop}$ consisting of those families of predicates which are rounded and satisfy the one appropriate form of the triangle inequality. Thinking of these predicates as half of a binary relation, we will write them as $(u, \epsilon) \mapsto (\diamondsuit \approx_\epsilon u)$, with the symbol $\diamondsuit$ referring to the whole relation. Now we can write A precisely as

$$\begin{aligned} A :\equiv \Bigg\{ \diamondsuit : \mathbb{R}_\mathsf{c} \to \mathbb{Q}_+ \to \mathsf{Prop} \;\Bigg|\; & \Big(\forall(u : \mathbb{R}_\mathsf{c}).\, \forall(\epsilon : \mathbb{Q}_+).\, \big((\diamondsuit \approx_\epsilon u) \Leftrightarrow \exists(\theta : \mathbb{Q}_+).\, (\diamondsuit \approx_{\epsilon-\theta} u)\big)\Big) \\ & \wedge \Big(\forall(u, v : \mathbb{R}_\mathsf{c}).\, \forall(\eta, \epsilon : \mathbb{Q}_+).\, (u \sim_\epsilon v) \to \\ & \big((\diamondsuit \approx_\eta u) \to (\diamondsuit \approx_{\eta+\epsilon} v)\big) \wedge \big((\diamondsuit \approx_\eta v) \to (\diamondsuit \approx_{\eta+\epsilon} u)\big)\Big) \Bigg\} \end{aligned}$$

As usual with subsets, we will use the same notation for an inhabitant of A and its first component $\diamondsuit$. As the family of relations required for $(\mathbb{R}_\mathsf{c}, \sim)$-recursion, we consider the following, which will ensure the other form of the triangle inequality:

$$(\diamondsuit \frown_\epsilon \heartsuit) :\equiv \forall(u : \mathbb{R}_\mathsf{c}).\, \forall(\eta : \mathbb{Q}_+).\, ((\diamondsuit \approx_\eta u) \to (\heartsuit \approx_{\epsilon+\eta} u)) \wedge ((\heartsuit \approx_\eta u) \to (\diamondsuit \approx_{\epsilon+\eta} u)).$$

We observe that these relations are separated. For assuming $\forall(\epsilon : \mathbb{Q}_+).\, (\diamondsuit \frown_\epsilon \heartsuit)$, to show $\diamondsuit = \heartsuit$ it suffices to show $(\diamondsuit \approx_\epsilon u) \Leftrightarrow (\heartsuit \approx_\epsilon u)$ for all $u : \mathbb{R}_\mathsf{c}$.

But $\diamondsuit \approx_\epsilon u$ implies $\diamondsuit \approx_{\epsilon-\theta} u$ for some θ, by roundedness, which together with $\diamondsuit \frown_\epsilon \heartsuit$ implies $\heartsuit \approx_\epsilon u$; and the converse is identical.

Now the first two data the recursion principle requires are the following.

- For any $q : \mathbb{Q}$, we must give an element of A, which we denote $(\mathsf{rat}(q) \approx_{(-)} -)$.
- For any Cauchy approximation x, if we assume defined a function $\mathbb{Q}_+ \to A$, which we will denote by $\epsilon \mapsto (x_\epsilon \approx_{(-)} -)$, with the property that

$$\forall (u : \mathbb{R}_\mathsf{c}).\, \forall (\delta, \epsilon, \eta : \mathbb{Q}_+).\, (x_\delta \approx_\eta u) \to (x_\epsilon \approx_{\eta+\delta+\epsilon} u), \tag{11.3.13}$$

  we must give an element of A, which we write as $(\mathsf{lim}(x) \approx_{(-)} -)$.

In both cases, we give the required definition by using a nested $(\mathbb{R}_\mathsf{c}, \sim)$-recursion, with codomain the subset of $\mathbb{Q}_+ \to \mathsf{Prop}$ consisting of rounded families of mere propositions. Thinking of these propositions as zero halves of a binary relation, we will write them as $\epsilon \mapsto (\bullet \approx_\epsilon \triangle)$, with the symbol $\triangle$ referring to the whole family. Now we can write the codomain of these inner recursions precisely as

$$C :\equiv \left\{ \triangle : \mathbb{Q}_+ \to \mathsf{Prop} \;\middle|\; \forall (\epsilon : \mathbb{Q}_+).\, \Big((\bullet \approx_\epsilon \triangle) \Leftrightarrow \exists (\theta : \mathbb{Q}_+).\, (\bullet \approx_{\epsilon-\theta} \triangle) \Big) \right\}$$

We take the required family of relations to be the remnant of the triangle inequality:

$$(\triangle \smile_\epsilon \square) :\equiv \forall (\eta : \mathbb{Q}_+).\, ((\bullet \approx_\eta \triangle) \to (\bullet \approx_{\epsilon+\eta} \square)) \wedge ((\bullet \approx_\eta \square) \to (\bullet \approx_{\epsilon+\eta} \triangle)).$$

These relations are separated by the same argument as for $\frown$, using roundedness of all elements of C.

Note that if such an inner recursion succeeds, it will yield a family of predicates $\diamondsuit : \mathbb{R}_\mathsf{c} \to \mathbb{Q}_+ \to \mathsf{Prop}$ which are rounded (since their image in $\mathbb{Q}_+ \to \mathsf{Prop}$ lies in C) and satisfy

$$\forall (u, v : \mathbb{R}_\mathsf{c}).\, \forall (\epsilon : \mathbb{Q}_+).\, (u \sim_\epsilon v) \to \big((\diamondsuit \approx_{(-)} u) \smile_\epsilon (\diamondsuit \approx_{(-)} u) \big).$$

Expanding out the definition of $\smile$, this yields precisely the third condition for $\diamondsuit$ to belong to A; thus it is exactly what we need.

It is at this point that we can give the definitions (11.3.6)–(11.3.9), as the first two clauses of each of the two inner recursions, corresponding to rational points and limits. In each case, we must verify that the relation is rounded and hence lies in C. In the rational-rational case (11.3.6) this is clear, while in the other cases it follows from an inductive hypothesis. (In (11.3.7) the relevant inductive hypothesis is that $(\mathsf{rat}(q) \approx_{(-)} y_\delta) : C$, while in (11.3.8) and (11.3.9) it is that $(x_\delta \approx_{(-)} -) : A$.)

The remaining data of the sub-recursions consist of showing that (11.3.6)–(11.3.9) satisfy the triangle inequality on the right with respect to the constructors of $\sim$. There are eight cases — four in each sub-recursion — corresponding to the eight possible ways that u, v, and w in (11.3.11) can be chosen to be rational points or limits. First we consider the cases when u is $\mathsf{rat}(q)$.

(i) Assuming $\mathsf{rat}(q) \approx_\phi \mathsf{rat}(r)$ and $-\epsilon < |r-s| < \epsilon$, we must show $\mathsf{rat}(q) \approx_{\phi+\epsilon} \mathsf{rat}(s)$. But by definition of $\approx$, this reduces to the triangle inequality for rational numbers.

(ii) We assume $\phi, \epsilon, \delta : \mathbb{Q}_+$ such that $\mathsf{rat}(q) \approx_\phi \mathsf{rat}(r)$ and $\mathsf{rat}(r) \sim_{\epsilon-\delta} y_\delta$, and inductively that

$$\forall(\psi : \mathbb{Q}_+).\, (\mathsf{rat}(q) \approx_\psi \mathsf{rat}(r)) \to (\mathsf{rat}(q) \approx_{\psi+\epsilon-\delta} y_\delta). \tag{11.3.14}$$

We assume also that $\psi, \delta \mapsto (\mathsf{rat}(q) \approx_\psi y_\delta)$ is a Cauchy approximation with respect to $\smile$, i.e.

$$\forall(\psi, \xi, \zeta : \mathbb{Q}_+).\, (\mathsf{rat}(q) \approx_\psi y_\xi) \to (\mathsf{rat}(q) \approx_{\psi+\xi+\zeta} y_\zeta), \tag{11.3.15}$$

although we do not need this assumption in this case. Indeed, (11.3.14) with $\psi :\equiv \phi$ yields immediately $\mathsf{rat}(q) \approx_{\phi+\epsilon-\delta} y_\delta$, and hence $\mathsf{rat}(q) \approx_{\phi+\epsilon} \mathsf{lim}(y)$ by definition of $\approx$.

(iii) We assume $\phi, \epsilon, \delta : \mathbb{Q}_+$ such that $\mathsf{rat}(q) \approx_\phi \mathsf{lim}(y)$ and $y_\delta \sim_{\epsilon-\delta} \mathsf{rat}(r)$, and inductively that

$$\forall(\psi : \mathbb{Q}_+).\, (\mathsf{rat}(q) \approx_\psi y_\delta) \to (\mathsf{rat}(q) \approx_{\psi+\epsilon-\delta} \mathsf{rat}(r)). \tag{11.3.16}$$

$$\forall(\psi, \xi, \zeta : \mathbb{Q}_+).\, (\mathsf{rat}(q) \approx_\psi y_\xi) \to (\mathsf{rat}(q) \approx_{\psi+\xi+\zeta} y_\zeta). \tag{11.3.17}$$

By definition, $\mathsf{rat}(q) \approx_\phi \mathsf{lim}(y)$ means that we have $\theta : \mathbb{Q}_+$ with $\mathsf{rat}(q) \approx_{\phi-\theta} y_\theta$. By assumption (11.3.17), therefore, we have also $\mathsf{rat}(q) \approx_{\phi+\delta} y_\delta$, and then by (11.3.16) it follows that $\mathsf{rat}(q) \approx_{\phi+\epsilon} \mathsf{rat}(r)$, as desired.

(iv) We assume $\phi, \epsilon, \delta, \eta : \mathbb{Q}_+$ such that $\mathsf{rat}(q) \approx_\phi \mathsf{lim}(y)$ and $y_\delta \sim_{\epsilon-\delta-\eta} z_\eta$, and inductively that

$$\forall(\psi : \mathbb{Q}_+).\, (\mathsf{rat}(q) \approx_\psi y_\delta) \to (\mathsf{rat}(q) \approx_{\psi+\epsilon-\delta-\eta} z_\eta), \tag{11.3.18}$$

$$\forall(\psi, \xi, \zeta : \mathbb{Q}_+).\, (\mathsf{rat}(q) \approx_\psi y_\xi) \to (\mathsf{rat}(q) \approx_{\psi+\xi+\zeta} y_\zeta), \tag{11.3.19}$$

$$\forall(\psi, \xi, \zeta : \mathbb{Q}_+).\, (\mathsf{rat}(q) \approx_\psi z_\xi) \to (\mathsf{rat}(q) \approx_{\psi+\xi+\zeta} z_\zeta). \tag{11.3.20}$$

Again, $\mathsf{rat}(q) \approx_\phi \mathsf{lim}(y)$ means we have $\xi : \mathbb{Q}_+$ with $\mathsf{rat}(q) \approx_{\phi-\xi} y_\xi$, while (11.3.19) then implies $\mathsf{rat}(q) \approx_{\phi+\delta} y_\delta$ and (11.3.18) implies $\mathsf{rat}(q) \approx_{\phi+\epsilon-\eta} z_\eta$. But by definition of $\approx$, this implies $\mathsf{rat}(q) \approx_{\phi+\epsilon} \mathsf{lim}(z)$ as desired.

Now we move on to the cases when u is $\mathsf{lim}(x)$, with x a Cauchy approximation. In this case, the ambient inductive hypothesis of the definition of $(\mathsf{lim}(x) \approx_{(-)} -) : A$ is that we have $(x_\delta \approx_{(-)} -) : A$, so that in addition to being rounded they satisfy the triangle inequality on the right.

(v) Assuming $\mathsf{lim}(x) \approx_\phi \mathsf{rat}(r)$ and $-\epsilon < |r-s| < \epsilon$, we must show $\mathsf{lim}(x) \approx_{\phi+\epsilon} \mathsf{rat}(s)$. By definition of $\approx$, the former means $x_\delta \approx_{\phi-\delta} \mathsf{rat}(r)$, so that above triangle inequality implies $x_\delta \approx_{\epsilon+\phi-\delta} \mathsf{rat}(s)$, hence $\mathsf{lim}(x) \approx_{\phi+\epsilon} \mathsf{rat}(s)$ as desired.

(vi) We assume $\phi, \epsilon, \delta : \mathbb{Q}_+$ such that $\mathsf{lim}(x) \approx_\phi \mathsf{rat}(r)$ and $\mathsf{rat}(r) \sim_{\epsilon-\delta} y_\delta$, and two unneeded inductive hypotheses. By definition, we have $\eta : \mathbb{Q}_+$ such that $x_\eta \approx_{\phi-\eta} \mathsf{rat}(r)$, so the inductive triangle inequality gives $x_\eta \approx_{\phi+\epsilon-\eta-\delta} y_\delta$. The definition of $\approx$ then immediately yields $\mathsf{lim}(x) \approx_{\phi+\epsilon} \mathsf{lim}(y)$.

(vii) We assume $\phi, \epsilon, \delta : \mathbb{Q}_+$ such that $\mathsf{lim}(x) \approx_\phi \mathsf{lim}(y)$ and $y_\delta \sim_{\epsilon-\delta} \mathsf{rat}(r)$, and two unneeded inductive hypotheses. By definition we have $\xi, \theta : \mathbb{Q}_+$ such that $x_\xi \approx_{\phi-\xi-\theta} y_\theta$. Since y is a Cauchy approximation, we have $y_\theta \sim_{\theta+\delta} y_\delta$, so the inductive triangle inequality gives $x_\xi \approx_{\phi+\delta-\xi} y_\delta$ and then $x_\xi \sim_{\phi+\epsilon-\xi} \mathsf{rat}(r)$. The definition of $\approx$ then gives $\mathsf{lim}(x) \approx_{\phi+\epsilon} \mathsf{rat}(r)$, as desired.

(viii) Finally, we assume $\phi, \epsilon, \delta, \eta : \mathbb{Q}_+$ such that $\mathsf{lim}(x) \approx_\phi \mathsf{lim}(y)$ and $y_\delta \sim_{\epsilon-\delta-\eta} z_\eta$. Then as before we have $\xi, \theta : \mathbb{Q}_+$ with $x_\xi \approx_{\phi-\xi-\theta} y_\theta$, and two applications of the triangle inequality suffices as before.

This completes the two inner recursions, and thus the definitions of the families of relations $(\mathsf{rat}(q) \approx_{(-)} -)$ and $(\mathsf{lim}(x) \approx_{(-)} -)$. Since all are elements of A, they are rounded and satisfy the triangle inequality on the right with respect to $\sim$. What remains is to verify the conditions relating to $\frown$, which is to say that these relations satisfy the triangle inequality on the *left* with respect to the constructors of $\sim$. The four cases correspond to the four choices of rational or limit points for u and v in (11.3.12), and since they are all mere propositions, we may apply $\mathbb{R}_\mathsf{c}$-induction and assume that w is also either rational or a limit. This yields another eight cases, whose proofs are essentially identical to those just given; so we will not subject the reader to them. □